

Week 2 Discussion

🔔 This Week's Topic: Lines, Arrows, and Curves

Our lecture moved from demand's own past to what demand moves with: Demand = $a + b \times X$. This reading asks two things of that line. Is it an **arrow**, so that changing X changes demand? And is it even a line, or a **log-log** curve with a constant elasticity, or a **U-shape** with a turning point? Read it, then find a real pair of your own. *Example* shows a top-scoring share. *Additional* (zip) has a no-code Colab guide and a **bonus question worth up to 10%**.

Reading: Lines, Arrows, and Curves

In lecture we put a number on a relationship. Demand = $a + b \times X$: a is where the line starts, b is how much demand changes when X rises by one unit, and the residual is the forecast's miss for one case (Jacobs & Chase, Chapter 18). The line is placed so that the squared misses add up to the smallest total. That part is arithmetic. What the line means is not. This reading separates three questions that the arithmetic cannot answer: whether the line is an arrow, which curve hides behind it, and what a planner may do with each.

A pattern can predict; only a cause justifies a change

Two things are related when knowing one tells you something about the other. The correlation r , a number between -1 and $+1$, says how tightly the dots hug a straight line. A strong r is enough to forecast with. If a top opponent is coming, expect a bigger crowd and staff more gates. You do not need to know why the pattern holds, only that it will hold next week.

A cause is a stronger claim. X causes demand when changing X , and only X , changes demand. Price cuts, promotions, and extra checkout lanes pay off only if the lever itself moves demand. So the same line answers two different questions. What will happen? A pattern is enough. What will happen if we do this? You need an arrow.

Three questions before you call a line an arrow



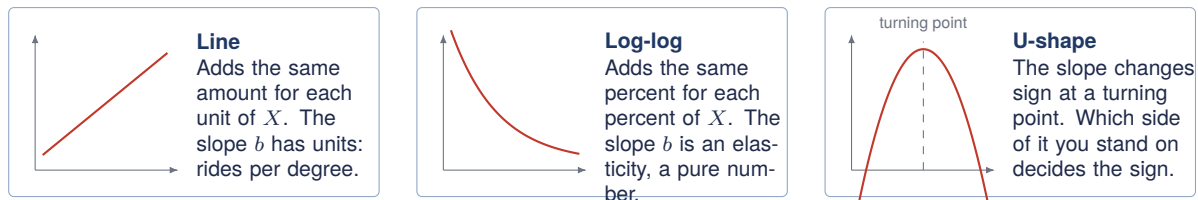
1. Could something else move both? Searches for air conditioning and for barbecue rise and fall together every year. Neither causes the other. Summer moves both. A common shock makes two series look linked when each is only linked to a third thing. The usual suspects are the season, the economy, a holiday, and a lockdown.

2. Could the arrow run the other way? Stores that hold more stock sell more. That is true in the data and useless as a lever, because the planner set the stock to match the sales forecast. The arrow

runs from expected sales to stock, not from stock to sales. Prices work the same way. A retailer cuts the avocado price in the week before the Super Bowl because it expects to sell more, so part of any line from price to volume is the retailer reacting to demand.

3. Am I seeing all the cases? A line is fit on the cases you can see. Sales data record what sold, not what was wanted: a store that ran out at two in the afternoon logs the sales it made, not the demand it had (Conrad, 1976). New-home sales count the buyers who closed, not the buyers a higher rate pushed out. And the line is only fit over the range you have seen. Outside that range it can bend, and the bike data below show how.

Two curves that hide behind a line



A straight line says one more unit of X always adds b units of demand. Two other shapes are common, and each changes what b means.

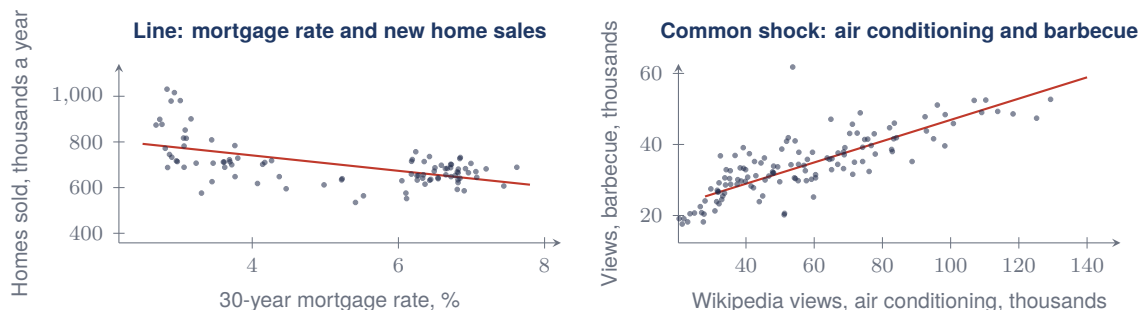
Log-log: the same percent, not the same amount. Take logarithms of both sides: $\ln(\text{Demand}) = a + b \ln(X)$. Now b is an *elasticity*. When X rises by one percent, demand changes by b percent, at any level of X . Price elasticity is the planner's version. An elasticity of -0.6 says a 10% price cut lifts volume by about 6%, whether the price is 80 cents or a dollar and a half. The line in levels cannot say that, because ten cents off a cheap item is a bigger cut than ten cents off a dear one. Elasticities are pure numbers, so they travel: you can compare avocados with gasoline.

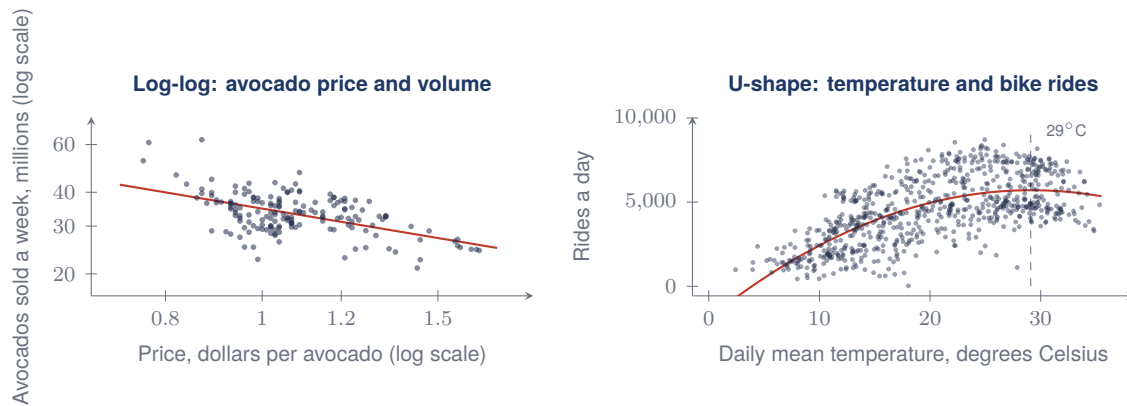
U-shape: the sign flips at a turning point. Add a squared term: $\text{Demand} = a + b_1X + b_2X^2$. The slope is now $b_1 + 2b_2X$. It changes as X grows and crosses zero at the turning point $X^* = -b_1/(2b_2)$. When $b_2 < 0$ the curve rises and then falls, an inverted U. Bike rides rise with temperature until the high twenties and fall when the day is hot. A planner who fits a straight line to such data gets a slope that is right on average and wrong at both ends. The question to ask is which side of the turning point you stand on.

The step tells the shapes apart, as it did in Week 1. A line adds a constant amount for each unit of X . A log-log curve adds a constant percent for each percent of X . A U-shape adds a shrinking amount, then a negative one.

Four relationships in real data

Each idea above shows up in real demand. The four pairs below are drawn as one dot per period with the fitted line or curve.





Sources: Freddie Mac and U.S. Census Bureau via FRED, monthly, January 2019 to August 2026; Wikimedia Pageviews API, monthly user views, 2016 to September 2026; Hass Avocado Board retail scan data, Total U.S., conventional, weekly, 2015 to March 2018; UCI Machine Learning Repository, Capital Bikeshare, daily, 2011 to 2012.

The fit says different things in the four panels. The table gives each one's correlation, its fitted form, what one step up in X does, and whether the line is an arrow.

Shape and pair	r	Fitted form	One step up in X	Arrow?
Line: mortgage rate, home sales	-0.57	$876 - 34 \times \text{rate}$	+1 point: -34 thousand homes a year	Yes, rate to sales
Common shock: air conditioning, barbecue	0.82	$17.0 + 0.30 \times \text{AC}$	+10 thousand AC views: +3 thousand barbecue	No, summer moves both
Log-log: avocado price, volume	-0.54	elasticity -0.62	+10% price: about -6% volume	Both ways
U-shape: temperature, bike rides	0.63	peak at 29°C	+1°: +343 at 10°, -51 at 32°	Yes, weather to rides

Line, mortgage rates. From 2019 to 2026 the 30-year rate ran from 2.7% in December 2020 to 7.6% in October 2023, and new single-family home sales ran from 1.03 million a year in October 2020 down to 535 thousand in July 2022. Each extra point on the rate goes with about 34 thousand fewer homes a year, and $r = -0.57$. The arrow mostly runs from the rate to sales: a higher rate raises the monthly payment, and buyers drop out. The Fed raises rates when the economy runs hot, which also lifts sales, so the common shock pushes the line toward zero and the true effect is if anything larger. And the cases are incomplete in a telling way. In 2023 and 2024 builders paid to buy down their buyers' rates, so the rate a new-home buyer actually paid sat below the posted rate. New-home sales held near 670 thousand a year while existing-home sales fell to their lowest level in decades.

Common shock, air conditioning and barbecue. Monthly views of the two Wikipedia articles move together, $r = 0.82$. Air conditioning draws 45 thousand views in a typical January and 84 thousand in a July; barbecue draws 31 thousand and 41 thousand. Nobody buys a grill because the air conditioner is running. Summer moves both. For a store that sells both, the pattern is a fine forecast signal and a useless lever: a sale on grills will not sell one air conditioner, and the pattern itself breaks in any year the summer comes late.

Log-log, avocados. Across 169 weeks from 2015 to early 2018, the U.S. average price of a conventional Hass avocado ran from 76 cents to \$1.65 and weekly volume from 21 million to 63 million. In logarithms the slope is -0.62: a 10% higher price goes with about 6% lower volume. The arrow runs both ways. A short crop in 2017 pushed the price to its high of \$1.65 in October and volume down to 24 million, a supply shock that moved both. And the three biggest weeks in the data are

the three weeks ending on Super Bowl Sunday, 61 to 63 million avocados at 77 to 87 cents, when retailers cut the price because they expect guacamole demand. So -0.62 mixes the shopper's response to the price with the retailer's response to the shopper. The elasticity you need for a pricing decision is the first part only.

U-shape, bike rides. Over 731 days in 2011 and 2012, Capital Bikeshare in Washington, DC, logged about 1,700 rides on a day below 10 degrees Celsius, 5,600 between 20 and 25, and 5,400 above 30. The fitted curve peaks at 29 degrees. Its slope is +343 rides per degree at 10 degrees, +164 at 20, and -51 at 32. A straight line through the same dots says +162 rides per degree everywhere, which is right in the middle of the range and wrong at both ends. The arrow runs from the weather to the rides, and nobody can pull it; the planner's job is to read tomorrow's forecast and put the bikes where the morning will want them.

From line to action

The planner's problem is that one line answers three different questions. Before you forecast with it, ask whether the pattern will still hold next period; a common shock that goes away takes the pattern with it. Before you pull a lever, ask whether the arrow is real and which way it points; if the arrow runs from sales to stock, more stock will not bring more sales. And before you extend the line, ask which curve you are on. A log-log slope gives a percent answer you can carry to any price. A U-shape says the same lever helps on one side of the turning point and hurts on the other, and that the data you have may sit entirely on one side.

Your Share

Due: Thursday, October 15, 2026 Post to Discussion Week 2

Peer review: Score the share you drew, following the course instructions.

After reading the introduction, find a real pair, demand and something it moves with, and share your reading of it. A written post, a chart, a one-page slide, or a short video all work. It must include:

- (1) what makes a line an arrow, and the two curves that can hide behind a line, in your own words, in two or three sentences;
- (2) your pair, with a link, shown as a scatter plot or a short run of numbers. Ask the three questions first, say whether you believe the arrow and which way it points, then say which shape the dots follow and check whether the data agree; and
- (3) your call as a planner: would you forecast with this relationship, pull it as a lever, or neither, and what evidence would change your mind.

Look for the line everyone reads as an arrow, or the lever everyone pulls that the data say does nothing. Find the price everyone calls elastic, or the relationship that bends once the data run far enough. That is the share worth reading.